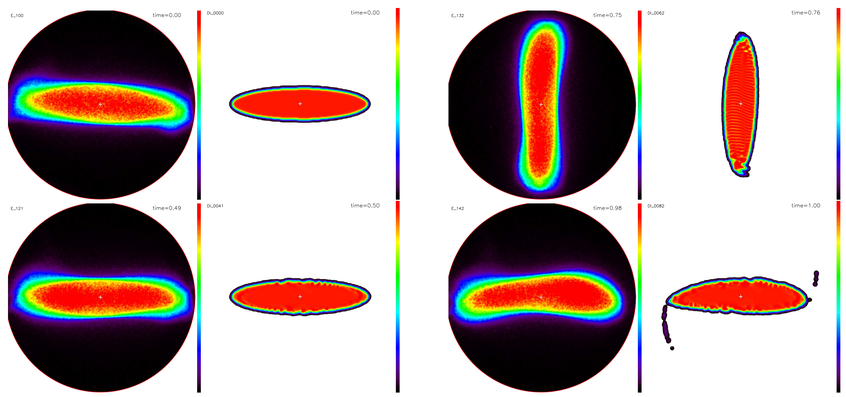
Revisiting instabilities in highly elliptical vortices

L. F. Rossi*, T. B. Mitchell† and X. J. Wang†

*Department of Mathematical Sciences †Department of Physics and Astronomy

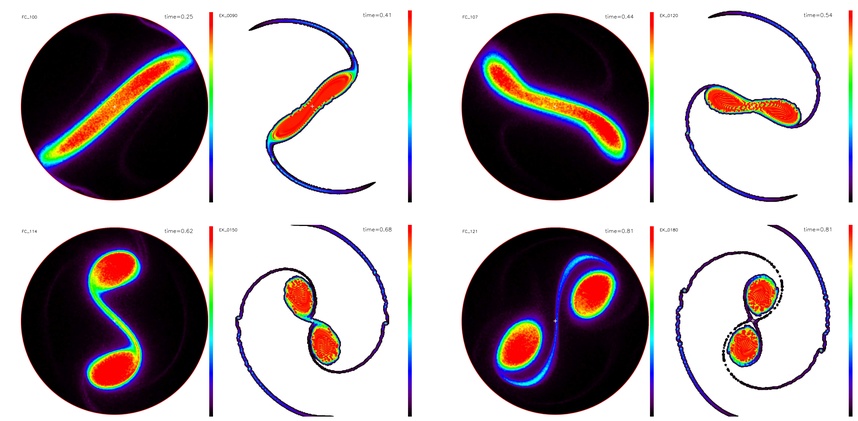
University of Delaware – Newark DE, USA

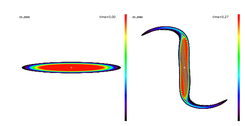
The growth of $m = 3$ perturbations in elliptical vortices.



After an elliptical vortex with aspect ratio of 4.3 is perturbed with an $m = 3$ disturbance, the perturbation grows exponentially, accompanied by filamentation. Filaments in the electron vortex data (left) are lost to the conducting circular boundary of the trap. This boundary also suppresses the growth rate of the m=3 modes slightly when compared with the free-space computational data (right).

Growth of the $m = 4$ and $m = 6$ modes with and without a circular boundary.





Highly elliptical Kirchhoff vortices with an aspect ratio of $a/b \sim 9$ were predicted in 1893 to be unstable to $m = 3$, 4, 5 and 6 modes [3]. The images show $m = 4$ and 6 modes growing from small perturbations, and saturating with two isolated vortical structures. Times are normalized to rotation period. The circular boundary increases the rotation rate in the electron experiment data. The early evolution from the open domain vortex method simulation data is shown to the left.

Synopsis

The dynamics of Kirchhoff vortices, elliptical patches of uniform vorticity surrounded by irrotational fluid, have a long history of scientific exploration using theoretical, experimental and computational techniques. In 1893, Love [3] determined that isolated Kirchhoff vortices in free space are unstable to azimuthal perturbations when the aspect ratio is greater than 3. The key mechanisms involved are rotation of the vortex, the growth of disturbances along the boundary called Kelvin modes and filamentation wherein vorticity is ejected from the bulk structure as studied by Melander, Zabusky, McWilliams [4] and many others. In this poster, we explore the linear growth and nonlinear evolution of very small perturbations of highly elliptical vortex patches both experimentally and computationally. This study introduces a few twists on the classic topic. First, the electron vortex apparatus imposes a cylindrical free-slip boundary condition that is distinct from those studied for isolated vortices. Second, we can control the width and structure of the transition between the uniform vorticity in the core region and the irrotational fluid outside the core. While it is convenient to treat the gradual transition as a perturbation to a Kirchhoff vortex, we find that it has a profound effect on initial linear growth of unstable modes as predicted by Love [3] and explored by Dritchell [1]. When the gradual transition at the edge expels filaments, the filaments can excite or suppress the Love instabilities. Our investigation is distinct from axisymmetrization studies because the filamentation is fast compared to the growth of Love modes. In this study, filamentation is controlled by the cylindrical boundary in the electron column experiments and by the imposed sharp edge in the vortex method computations. These instabilities also can be triggered by slightly asymmetric vortex mergers.

References

[1] David G. Dritschel. The nonlinear evoluation of rotating configurations of uniform vorticity. *J. Fluid Mech.*, 172:157–182, 1986.

[2] A. Leonard. Vortex methods for flow simulation. *J. Comp. Phys.*, 37:289–335, 1980.

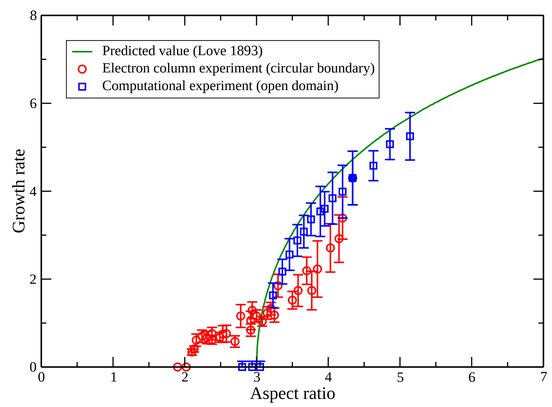
[3] A. E. H. Love. On the stability of certain vortex motions. *Proc. London Math. Soc.*, 25:18–42, 1893.

[4] M. V. Melander, J. C. McWilliams, and N. J. Zabusky. Axisymmetrization and vorticity-gradient intensification of an isolated two-dimensional vortex through filamentation. *J. Fluid Mech.*, 178:137–159, 1987.

[5] T B Mitchell and C F Driscoll. Electron vortex orbits and merger. *Phys. Fluids*, 8(7):1828–1841, 1996.

[6] Louis F. Rossi. Achieving high-order convergence rates with deforming basis functions. *SIAM J. Sci. Comput.*, 26(3):885–906, 2005.

Stability of the $m = 3$ Love mode.

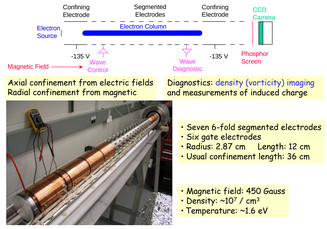


The simulation growth rates agree well with the rates predicted by Love for a Kirchhoff vortex in an open domain. The experiment is different in that there is a circular boundary, and the vorticity profile is not infinitely steep. Possibly because of these effects, with the electron vortices we observe slower growth rates and a lower aspect ratio instability threshold for the $m = 3$ mode. The data point indicated by solid symbol is from the simulation shown in the top left panel.

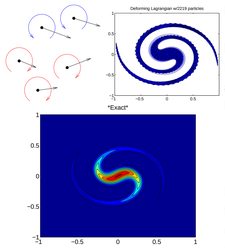
Techniques

Experimental methods

Electron vortices are pure electron plasma columns trapped within hollow conducting cylinders [5]. Negative voltages on the end electrodes provide axial confinement, and a uniform axial magnetic field provides radial confinement. A rapid axial bounce motion averages over z variations, allowing a 2D description of the system. The r, θ flow of the electrons is well described by 2D $\vec{E} \times \vec{B}$ dynamics, with an equation of motion which is isomorphic to the 2D Euler equation for an inviscid fluid of uniform density within a circular boundary. The vorticity of the electron flow is proportional to the electron line density, which is directly imaged in the experiment. Electron vortices have simple free-slip boundary conditions and low dissipation rates compared to vortices in conventional fluids.



Computational methods



The vortex methods have a long and respected history as a naturally adaptive method for computing vorticity dominated flows. These techniques represent the vorticity field of a fluid flow as a linear combination of moving, localized basis functions. Each basis function can be interpreted as a small patch of vorticity which influences all the other basis functions (see schematic at top left). Collections of these basis functions or blobs, such as those shown at the top right (blob positions) and bottom (sum of all basis function contributions), approximate the vorticity equation

$$\omega_t + \vec{u} \cdot \nabla \omega = \frac{1}{\mathrm{Re}} \nabla^2 \omega, \qquad \nabla \cdot \vec{u} = 0$$

by moving with the physical flow field calculated using the Biot-Savart law [2, 6]. For these computations the Reynolds number is considered infinite so the only dissipation is through numerical regularization. Initial conditions with sharp edges are generated using radial basis function projection methods.